

TRANSFORMER ARCHITECTURE: MULTI-HEAD SELF-ATTENTION MECHANISM



LAYMAN TERM

Imagine you have a magical talking book where every word in a sentence can look at all the other words at the same time to understand the story better. Multi-head self-attention is like having many pairs of special glasses, each helping a word see the story in a different way, making the magic super strong and smart!



DEFINITION

Multi-head self-attention is a component in transformers where multiple attention mechanisms run in parallel to capture varied relationships between elements in a sequence for contextual understanding.



WHEN TO USE

Use multi-head self-attention in tasks requiring understanding of context and relationships in sequences, like language translation, text generation, and speech processing, where capturing diverse word interactions is crucial.



HOW TO MAKE

- Start with an input sequence and convert words into vectors (embeddings).
- Create three matrices from these embeddings: Query (Q), Key (K), and Value (V).
- Split these matrices into multiple sets for multiple attention heads.
- Calculate attention scores for each head by comparing queries with keys.
- Apply softmax to normalize scores and multiply by values to get outputs.



TYPES

- Single-Head Attention: A basic attention mechanism focusing on one aspect.
- Multi-Head Attention: Multiple attention heads capturing different relationships.
- Scaled Dot-Product Attention: Attention using scaled dot products for more stable gradients.
- Masked Multi-Head Attention: Attention restricted to previous tokens, used in decoding to prevent future look ahead.
- Cross-Attention: Attention mechanism focusing on



POINTS TO PONDER

- Multi-head attention requires significant computational resources due to parallel processing.
- Choosing the number of heads is a trade-off between model performance and complexity.
- Attention mechanisms may struggle with very long sequences due to quadratic scaling of computations.
- Proper training and data are essential for the attention to learn meaningful relationships.
- Positional encoding is necessary as transformers lack inherent sequence order awareness.

- Concatenate outputs from all heads into one matrix.
- Apply a final linear transformation to combine all heads' outputs.
- Use this combined output for further processing in the model.

relationships between encoder and decoder inputs.



CONCLUSION

Multi-head self-attention allows transformers to learn complex relationships by attending to different parts of input simultaneously. Formula: $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$, where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$.



RECOMMENDED SOURCES

- <https://www.datacamp.com/tutorial/multi-head-attention-transformers>
- <https://www.geeksforgeeks.org/nlp/multi-head-attention-mechanism/>
- <https://mljourney.com/transformer-architecture-explained-for-beginners/>



BEST VIDEO GUIDE

RECOMMENDED CONTENT:

Multi-Head Self-Attention explained Microsoft Mechanics

LINK: [HTTPS://YOUTUBE.COM/WATCH?V=TDZIROAWH14](https://youtube.com/watch?v=TDZIROAWH14)